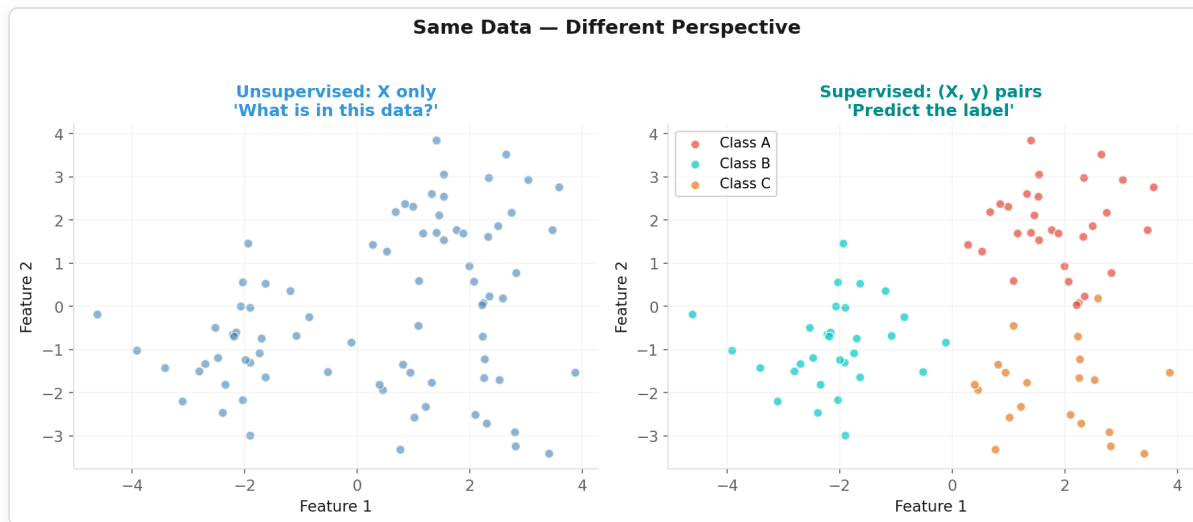


Introduction to Unsupervised Learning

Applied Machine Learning — Session 3, Chapter 1

The Key Difference



No labels. No "right answer." Just data.

Why Does Unlabeled Data Exist?

- Labeling is **expensive** (human annotation costs time + money)
- Labeling is **impossible** for future data
- Labels **don't exist yet** (discovery science: "are there disease subtypes?")
- We don't know what we're looking for

Most data in the world is unlabeled.

Supervised learning is the exception, not the rule.

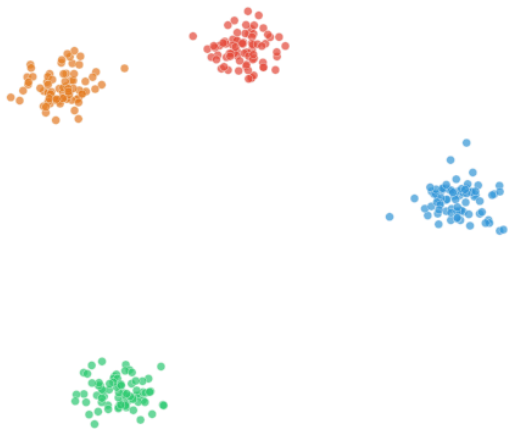
Analogy: sorting a pile of mail without rules — by sender, size, topic.

Different people sort differently. **Both can be valid.**

Find the Groups!

Reveal: your eyes clustered without labels — can an algorithm?

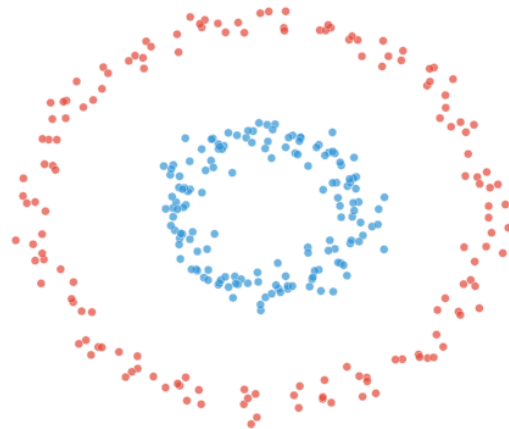
Blobs — the groups (labels we did NOT have)



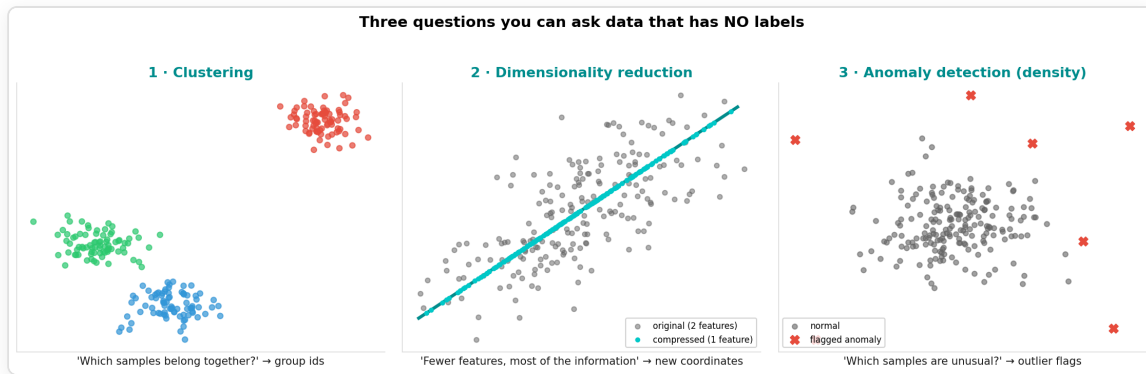
Moons — the groups (labels we did NOT have)



Circles — the groups (labels we did NOT have)



Three Questions You Can Ask Unlabeled Data



	Question	Output	Chapter
Clustering	Which samples belong together?	group id per sample	Ch08
Dimensionality reduction	Can I keep the information in fewer features?	new coordinates	Ch09
Anomaly detection	Which samples are unusual?	outlier flag	today, 1 line

Clustering & Dimensionality Reduction in One Line Each

```
from sklearn.cluster import KMeans
labels = KMeans(n_clusters=3, random_state=42).fit_predict(X)    # → array([0, 2, 1, 0, ...])
```

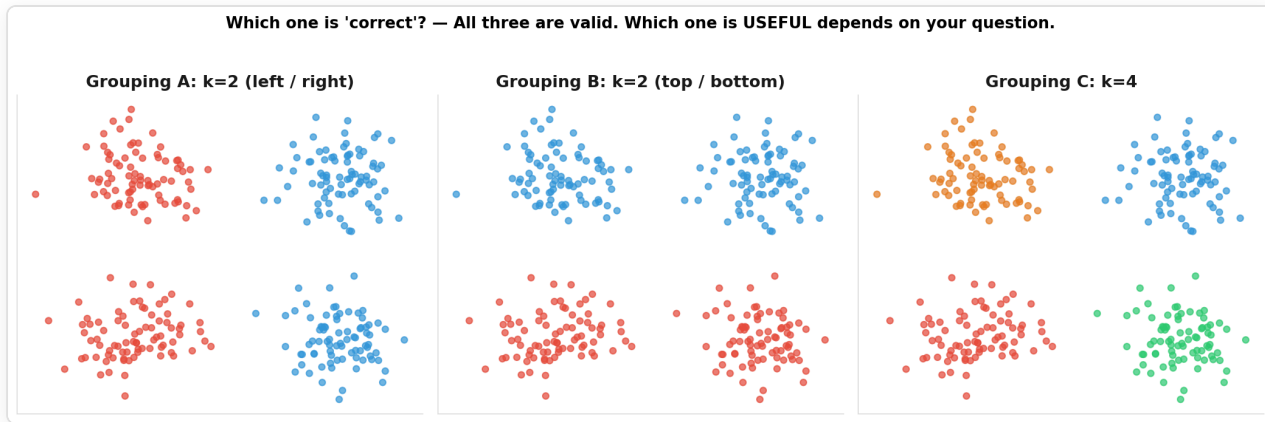
```
from sklearn.decomposition import PCA
X_2d = PCA(n_components=2).fit_transform(X)                      # → shape (n, 2) – plot it!
```

```
from sklearn.ensemble import IsolationForest
is_outlier = IsolationForest(random_state=42).fit_predict(X) == -1    # → True for unusual rows
```

Same sklearn grammar as always: `fit` → but there is **no** `y`.

The Evaluation Challenge

Without labels — how do we know if we did a good job?



Internal metrics (no labels): Silhouette score, inertia — "are groups compact and separated?"

External checks: compare with labels *if you happen to have some* (ARI) — a sanity check, not the goal

Domain sense: can a person explain each group? Does it help a downstream task?

Real-World Applications

Field	Application	Technique
Marketing	Customer segments	Clustering
Medicine	Disease subtypes from gene expression	Clustering
NLP	Topic discovery	Clustering
Finance / IT	Fraud, intrusion, sensor faults	Anomaly detection
Vision	Image compression, face "eigen-directions"	PCA
Any	Visualising 100-D data in 2-D	PCA / t-SNE / UMAP
Recommender systems	Latent user/item factors	Dim. reduction

Let's Explore Together

→ Open `02-examples/ch07_unsupervised_intro_examples.ipynb`

We will (~12 min):

1. Take Iris, **delete the labels**, look at what remains
2. Look at three kinds of "structure" (blobs, moons, circles)
3. **Anomaly detection in one line** (IsolationForest on customer data)
4. A 30-second teaser of the digits dataset (64 features) — full treatment in Ch09

Quick Check

Q1. A colleague says: "I ran K-Means with $k=3$ on our customers and got 3 groups, so we have 3 customer types." What is wrong with that sentence?

Q2. You have 10,000 unlabeled X-ray images and 200 with a radiologist's label. Which paradigm(s) can use which part?

Quick Check

Q1. A colleague says: "I ran K-Means with $k=3$ on our customers and got 3 groups, so we have 3 customer types." What is wrong with that sentence?

→ K-Means returns k groups **whatever** the data looks like. 3 groups came out because 3 was asked for. Check compactness/separation (Ch08) and whether the groups make business sense.

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→ Unsupervised (clustering, PCA, anomaly detection) can use all 10,200; supervised only the 200. Combining both (e.g. PCA/clusters as features, or "semi-supervised") is common in practice.

Key Takeaways

- Unsupervised = learning **without a `y` column** — the normal state of real data
- Three questions: **clustering, dimensionality reduction, anomaly detection**
- Output is a column we invent: group id / new coordinates / outlier flag
- Evaluation is hard: internal metrics + domain judgement; **clusters $\neq$ classes**
- Optional exercise: `03-exercises/ch07_unsupervised_intro_exercises.ipynb` (PCA + K-Means on digits, ~15 min, homework)

Next: Chapter 8

Clustering Techniques

"Let's make the machine find the groups."